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Cáin agus Custaim na hÉireann
Irish Tax and Customs

Investment Account Technical Group Forum
Meeting 2

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Agenda

- Recap on Forum Meeting 1
- Presentation of Revenue REST API Integration
- Questions
- Next steps

Note: Department of Finance designated email address for policy queries (which cannot be routed via the industry bodies) is investmentaccount@finance.gov.ie

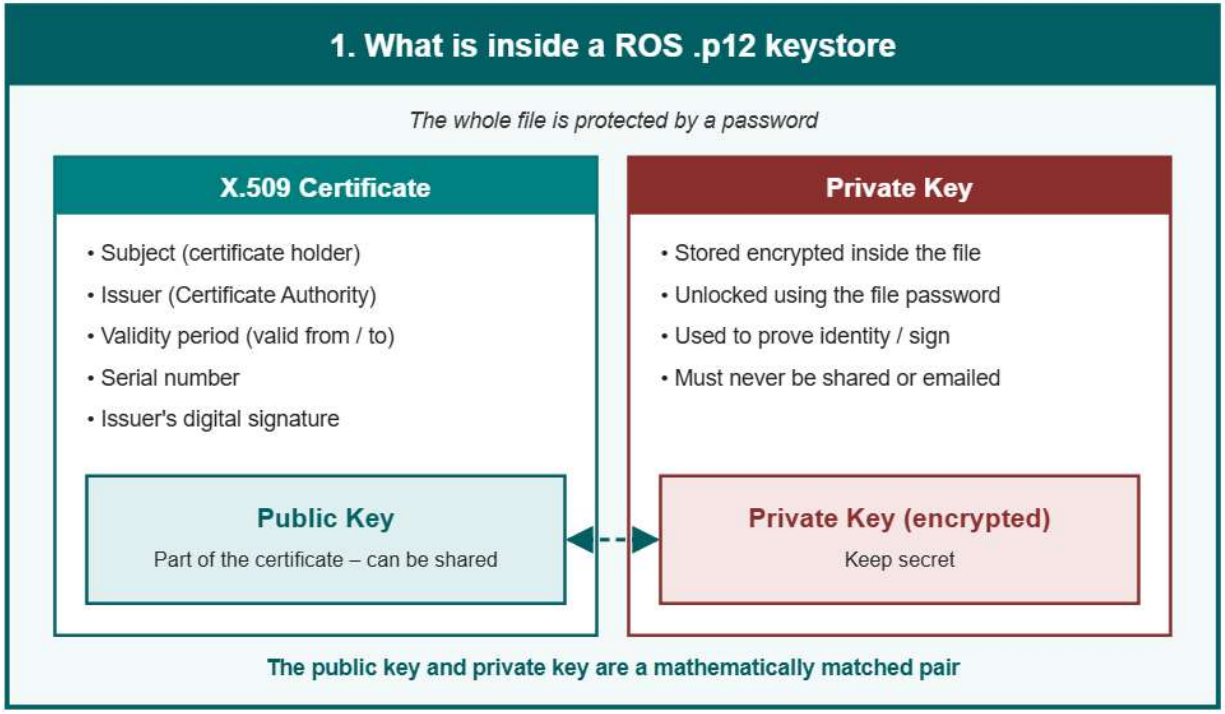
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Recap on Forum Meeting 1

- Eligibility
- Waitlist
- Authorisation
- Access to Schema

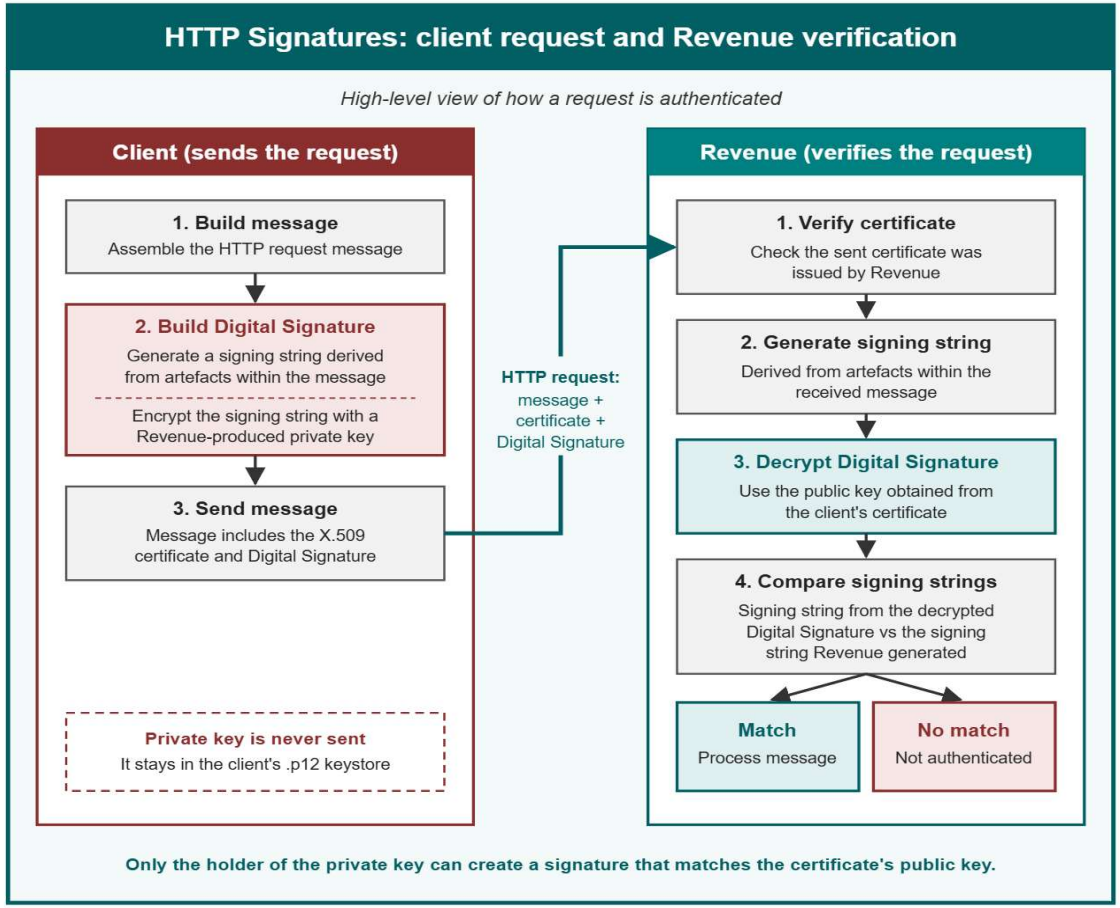
Integrating With Revenue REST API's

All Revenue web service endpoints that either accept or return confidential information must be digitally signed. The digital signature ensures the integrity of the message being sent. By signing the message we can ensure that the request data has not been tampered with. Messages must be signed using a ROS digital certificate that has been generated by Revenue. The contents of a ROS certificate are detailed below



HTTP Signatures Implementation

- Revenue leverages the HTTP signatures framework (RFC 9421) to facilitate digital signature creation and verification
- The HTTP signatures framework provides a simple and standard way for clients to sign HTTP requests.
- All required HTTP Signature details must be added to a 'Signature' HTTP header that should accompany every request
- All requests to the IIA API will require the following steps to be carried out by the integrator and Revenue



The Signature HTTP Header

- As mentioned on the previous slide, all required HTTP Signature details must be added to the ‘Signature’ HTTP header that accompanies every request to the IIA API. The ‘Signature’ HTTP header is comprised of a set of 4 fields.
- Below is an example of how a Signature HTTP Header is structured. Its made up of 4 fields

```
Signature:  keyId="MIICfzCCAeigAwIBAgIJ... // truncated",
            algorithm="rsa-sha512",
            headers="(request-target) host date digest",
            signature="GdUqDgy94Z8mSYUjr/rL6qrLX/jmudS... // truncated"
```

- keyId – Revenue issued X.509 Certificate
- algorithm - Digital signature algorithm used when generating the signature from the ‘signing string’
- headers – Ordered list of message artefacts used when generating the signature
- signature - Digital signature generated using the ‘headers’ field to form the ‘signing string’ along with the ‘algorithm’ field and the private key that generates the ‘signature’ field value.

The Signature HTTP Header details

- Once the required HTTP headers have been added to the HTTP request, we can begin to build the HTTP Signature header which is simply a concatenation of the following pieces of information

- **keyId** - X509 certificate from the ROS certificate p12 file expressed base 64 encoded byte array. This field is required.
  - Encode the Password used to open the keystore
  - Open the keystore
  - Get the Certificate from the keystore
  - Get Base64 Encoded Certificate As a String
- **algorithm**: The ‘algorithm’ field is used to specify the digital signature algorithm to use when generating the signature. Currently, Revenue expects this to be ‘rsa-sha512’. This field is required.
- **headers**: The 'headers' field specifies the list of headers used when generating the signature for the message. The parameter must be a lowercase, quoted list of HTTP header fields, separated by a single space character. The list order is important, and MUST be specified in the order the HTTP header field-value pairs are concatenated together when building the ‘signing string’.
- **signature**: Canonicalized ‘signing string’ which is signed with the ROS certificates private key using the specified digital signature algorithm from the ‘algorithm’ field. The 'signature' field is then base 64 encoded, converted to a String and concatenated with the ‘keyId’, ‘algorithm’ and ‘headers’ fields to form the Signature HTTP header.

Signature Field - How to build a ‘signing string’ 1

- Ensure all artefacts used in the generation of the ‘signing string’ are added to the HTTP request before generating the signing string. Each artefact should be specified in the 'headers’ portion of the ‘Signature’ HTTP header.
- Artefacts from the HTTP request that can be used in the headers field are outlined in the table below

Value	Mandatory
(request-target)	Yes
host	Yes
date	Yes
x-date	Yes, if date header cannot be added.
digest	Yes, if HTTP method is of type POST
content-type	No
content-length	No

Signature Field - How to build a ‘signing string’ 2

In order to generate the ‘signing string’, the implementer MUST use the values of each HTTP header/artefact defined in the 'headers' signature field. Values must be in the order they appear in the 'headers' field of the ‘Signature’ HTTP header. If an associated HTTP header defined in the ‘headers’ field list does not exist, it should be added to the HTTP request BEFORE attempting to construct the signing string.

The (request-target) header is built from 2 separate HTTP headers. It is generated by concatenating the lowercase ‘HTTP method’ HTTP header, an ASCII space, and the ‘request path’ HTTP header.

All other header field values are created by concatenating the lowercase header field name followed by an ASCII colon ':', an ASCII space ' ', and the header field value. Leading and trailing whitespace in the header field value MUST be omitted. If the header field is not the last value defined in the 'headers' signature field, then append an ASCII newline '\n’ that should look something like the example below

(request-target): get / individual-investment-account /v1/rest/handshake?softwareUsed=test&softwareVersion=1.0\n
host: softwaretestnextversion.ros.ie\n
date: Wed Jun 13 2018 11:37:48 GMT+0100 (GMT Daylight Time

Steps to build the ‘signature’ field

1. Generate the ‘signing string’
2. Encode the Password used to open the keystore
3. Open the keystore
4. Get private key From KeyStore
5. Sign string

- Once all 4 fields have been created and concatenated, the ‘Signature’ header should look similar to the example below

```
Signature: keyId="MIICfzCCAeigAwIBAgIJ... // truncated",
algorithm="rsa-sha512",
headers="(request-target) host date digest",
signature="GdUqDgy94Z8mSYUjr/rL6qrLX/jmudS... // truncated“
```

- Quick Demo

Questions

- Any questions?

Next Steps

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Provisional date for next meeting of Forum is 25.11.2026 at 10am via MicroSoft Teams